Project: AI-Driven Business

Introduction to the Project Exam for Business and Project Management

AI-Driven Business

Project Purpose

Bridges AI technical knowledge with management and strategic business thinking for engineering students.

Student Expectations

Students develop an AI company focusing on value, market feasibility, and strategic risk over algorithms.

Interdisciplinary Reasoning

Combines engineering logic, market analysis, and structured business planning for comprehensive understanding.

Real-World Skills

Prepares students to assess AI’s strategic implications and risks in innovation-driven industries.

Objectives of the Project

Designing AI Business Ventures

The project focuses on designing AI-driven business solutions that address real-world industry problems effectively.

Market and Competitive Analysis

Students analyze market potential and competitive dynamics to validate business viability and strategic positioning.

Risk and Regulatory Awareness

The project encourages reflection on long-term risks and understanding regulatory considerations in AI business contexts.

Structured Decision-Making

Emphasis is placed on clear reasoning, coherent assumptions, and translating AI capabilities into economic value.

Industry Grid and Rationale

Industry Selection Importance

Selecting a single industry is crucial as it directs all further project analysis and outcomes.

AI as Enabling Technology

AI drives innovation by addressing industry challenges, leveraging data, and adapting to regulatory and technological trends.

Rationale Components

Rationale includes industry challenges, data availability, regulations, and tech trends to justify AI application.

Industries

- Real Estate
- Cyber Security
- Creative Industries
- Quantum Computing
- Quantum Optics
- GIS
- Automotive
- Social Robotics
- Interior Design
- Music & Sound Design
- Education

Data and Information Sources

Sources for Evidence-Based Analysis

Formal Data Sources

Patents, trademarks, and regulations provide insight into technology maturity and market entry barriers.

Open and Informal Sources

Social media, newspapers, and public datasets capture user needs, trends, and unmet problems.

Professional Market Intelligence

Financial data and market insights from tools help analyze competitive dynamics and market sizing.

Triangulation of Sources

Cross-verifying data from multiple sources ensures robust and defensible evidence-based analysis.

Project Structure

Structure

1 – Problem/Solution Fit

2 - Competition analysis

3 - Business Model Canvas

4 - Market Estimation

5 - Market test

6 - Negative Back casting

1 - Problem–Solution and Product–Market Fit

Defining the Problem

Clearly articulate the specific problem and identify who experiences it and why current solutions fall short.

AI-Enabled Solution

Explain the role of AI in providing superior performance, scalability, or personalization in the solution.

Achieving Product–Market Fit

Identify the target customer segment and demonstrate why the solution meets their needs better than alternatives.

Logical Assumptions Foundation

Make all key assumptions visible and logically connected to support business and market analyses.

2 - Competitors and Positioning

Competitive Analysis

Identify direct and indirect competitors including startups, incumbents, and substitute solutions without AI.

Assessing Competitors

Evaluate competitors' value propositions, strengths, and weaknesses to understand market dynamics.

Strategic Positioning

Differentiate the venture through cost, performance, data access, compliance, or customer experience.

Validating Competitiveness

Demonstrate strategic awareness and validate the venture's ability to compete realistically in the market.

3 - Business Model Canvas

Core Components Overview

The canvas outlines customer segments, value propositions, channels, revenue streams, and cost structures clearly.

AI-related Assets Focus

Special focus on AI assets including data, models, and computational infrastructure is essential for modern ventures.

Assumptions and Interdependencies

Highlighting critical assumptions and interdependencies ensures clarity on operational risks and scalability.

Feasibility and Scalability Logic

Translates conceptual ideas into economic logic to evaluate feasibility and scalability effectively.

4 - Addressable Market Analysis

Market Size Components

Estimate TAM, SAM, and SOM to understand the full economic potential for the venture.

Estimation Methodology

Use top-down or bottom-up approaches with clear assumptions to ensure transparency and logic.

Investment Decision Support

Market analysis helps evaluate if the opportunity justifies investment and development efforts.

5 - Personas and Validation Strategy

User Personas Definition

Define at least two user personas with roles, contexts, goals, and frustrations to guide market testing.

Validation Strategy Methods

Outline validation methods such as interviews, surveys, MVPs, and pilot projects to test ideas.

Learning and Decision-Making

Identify success and failure signals to enable learning from users and support evidence-based decisions.

6 - Failure Analysis and Risk Mitigation

Negative Backcasting Method

Students assume project failure and trace backward to identify possible causes and risks.

Common Failure Causes

Failures include strategic misalignment, market adoption issues, technological and regulatory barriers, and AI-specific risks.

Early Warning and Mitigation

Discuss early warning signals and mitigation actions to proactively address risks and avoid failure.

Strategic Thinking Benefits

Disciplined pessimism enhances strategic thinking and highlights AI venture complexities in the market.

Assessment Dimensions

Evaluation Criteria

Projects are assessed for logical consistency, depth of analysis, and appropriate data use to ensure quality.

Transparency and Coherence

Transparency of assumptions and coherence between sections are critical for credible project assessment.

Validity and Credibility

Results should be relevant, credible and grounded in data.

Clear Presentation

Clear and structured presentation is essential to communicate complex ideas to managerial audiences.